

Quantum and non-signalling graph isomorphisms

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . If such an isomorphism exists, we say that G and H are *isomorphic*. This is a well-known problem in graph theory. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Introduction 1

Abstract

We introduce two novel non-relaxed all-to-all bipartite graph isomorphism problems. We show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from

Abstract

We show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from

³University College London, United Kingdom

²University of Bristol, United Kingdom

¹Universitat Politècnica de Catalunya, Spain

1 Introduction

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . If such an isomorphism exists, we say that G and H are *isomorphic*. This is a well-known problem in graph theory. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental

Quantum and non-signalling graph isomorphisms

In the game, the verifier samples a pair $(x_A, x_B) \in X_A \times X_B$ using the distribution π and sends $x_A \in X_A$ to Alice and $x_B \in X_B$ to Bob. Alice and Bob respond with $y_A \in Y_A$ and $y_B \in Y_B$ respectively. We say the players win the game if $V(x_A, x_B, y_A, y_B) = 1$.

The game is defined in terms of finite input sets X_A, X_B and finite output sets Y_A, Y_B for Alice and Bob respectively. A Boolean predicate $V: X_A \times X_B \times Y_A \times Y_B \rightarrow \{0, 1\}$ and a distribution π on $X_A \times X_B$.

A two-party nonlocal game includes a verifier and two players, Alice and Bob, that devise a cooperative strategy. The game is defined in terms of finite input sets X_A, X_B and finite output sets Y_A, Y_B for Alice and Bob respectively. A Boolean predicate $V: X_A \times X_B \times Y_A \times Y_B \rightarrow \{0, 1\}$ and a distribution π on $X_A \times X_B$.

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Vavatsiotis⁶

¹University of Bristol, United Kingdom
²University College London, United Kingdom
³Charles University (Czech Republic)
⁴University College London, United Kingdom and Shanghai Jiao Tong University, China
⁵University College London, United Kingdom and Shanghai Jiao Tong University, China
⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

1.1 Nonlocal games

Although the term “quantum isomorphism” might seem unmotivated at this point, as we will see in the next section, feasibility of (IQP^d) corresponds to a relaxation of graph isomorphism based on the standard quantum mechanics available in quantum mechanics.

Although the term “quantum isomorphism” might seem unmotivated at this point, as we will see in the next section, feasibility of (IQP^d) corresponds to a relaxation of graph isomorphism based on the standard quantum mechanics available in quantum mechanics.

Although the term “quantum isomorphism” might seem unmotivated at this point, as we will see in the next section, feasibility of (IQP^d) corresponds to a relaxation of graph isomorphism based on the standard quantum mechanics available in quantum mechanics.

As we will see in Section 5, the system (IQP^d) is feasible if and only if (ILP^d) is feasible. Thus the feasibility of (IQP^d) (or equivalently (ILP^d)) corresponds to a “robust” relaxation of the notion of graph isomorphism which we call quantum isomorphism (see Definition 1.1).

Note that for $d = 1$ the matrix $\mathcal{P}$ is exactly a permutation matrix.

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by the standard reduction of graph isomorphism to a system of linear equations. Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\phi: V(G) \rightarrow V(H)$ such that $\phi(u)$ is adjacent to $\phi(v)$ in H if and only if u is adjacent to v in G . What isomorphism exists? First, we consider (IQP) where we relax the scalar variables to be non-commuting operators. We say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in complexity theory. This defines an equivalence relation on graphs that has been studied in detail and characterized in many ways [27]. If there exists a doubly stochastic matrix D such that $AD = B^T A^T$, then G and H are fractional isomorphic, and write $G \cong_f H$. This naturally suggests the following linear

Result 2. For any two graphs G, H the following are equivalent:

$$(i) \ G \cong^q H;$$

(ii) $G \cong^p H$ for some $p \in \mathbb{N}$. The system Π is feasible for some $p \in \mathbb{N}$.
 (iii) $G \cong^p H$ for some $p \in \mathbb{N}$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity and logic. The Π problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in complexity theory.

1 Introduction

Given graphs G and H , a natural question is whether $G \cong H$. This question is central to the theory of graph isomorphism. In this paper, we consider the question of whether $G \cong H$ is decidable, as well as the question of whether $G \cong H$ is tractable.

Result 3. The complexity of the graph isomorphism problem is $\Theta(n^2)$.
 The tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable.

The main motivation for this paper is the question of whether $G \cong H$ is decidable. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that two graphs are isomorphic if and only if they are isomorphic in the tensor product and commuting frameworks.

3 Strategies for the (G, H) -isomorphism game

1 Introduction

$$(4) \quad p_S(y_A, y_B | x_A, x_B) > 0 \text{ and } V(x_A, x_B) = 0, \text{ when } \pi(x_A, x_B) = 0, \text{ when } \pi(x_A, x_B) = 0.$$

In particular, if we specialize (4) to the (G, H) -isomorphism game, we have that the correlation is undecidable. Our construction is related to the EGLSS reduction from inapproximability literature, as well as the CFI construction.

In a classical case, we consider both the tensor product and commuting frameworks for non-local games. We prove that non-signalling isomorphism games are both the only and the most general fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming constraints for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, as a by-product, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable.

In this section we introduce three families of strategies for the (G, H) -isomorphism game (classical, quantum and non-signalling). It is a question with fundamental importance whether two graphs are isomorphic. In this section we introduce three families of strategies for the (G, H) -isomorphism game (classical, quantum and non-signalling). It is a question with fundamental importance whether two graphs are isomorphic. In this section we introduce three families of strategies for the (G, H) -isomorphism game (classical, quantum and non-signalling). It is a question with fundamental importance whether two graphs are isomorphic.

Since we can consider the deterministic strategy corresponding to p_1 . A deterministic strategy amounts to a pair of functions $V : (G \cup V(H)) \rightarrow V(G \cup V(H))$, which map inputs to outputs for each of Alice and Bob respectively. Assuming the strategy is winning, we have that $f_A(x) = f_B(x)$ for all $x \in V(G) \cap V(H)$, and that $f_A(x) = f_B(x)$ for all $x \in V(G) \cap V(H)$.

Since we will refer to both of them as isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for non-local games. We prove that non-signalling isomorphism games are both the only and the most general fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming constraints for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, as a by-product, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable.

Since we will refer to both of them as isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for non-local games. We prove that non-signalling isomorphism games are both the only and the most general fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming constraints for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, as a by-product, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable.

The above shows that any winning deterministic strategy for the (G, H) -isomorphism game corresponds to a perfect strategy if and only if the restriction of f to $V(G) \cap V(H)$ is the inverse of the restriction of f to $V(G) \cap V(H)$. Since f is a bijection $\phi : V(G) \rightarrow V(H)$ such that $\phi(g)$ is adjacent to $\phi(h)$ if and only if g is adjacent to h . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance whether two graphs are isomorphic.

$$\sum_{B \in \mathcal{B}} d(B, A) = \sum_{B \in \mathcal{B}} d(A, B) \quad (8)$$

Albert Atserias¹, Laura Mancinska², David E. Roberson³, Robert Šámal⁴, Simone

and we see that this does not depend on Bob's input x_B . Similarly, the probability of Bob obtaining a particular outcome will not be dependent on Alice's input. This property of quantum correlations is known as the *non-signalling* property. Severini⁵ and Antonopoulos⁶ have shown that the condition $(\rho_{AB} | \rho_{A,A} \otimes \rho_{B,B})$ is non-signalling if

³University College London, United Kingdom
Charles University, Czech Republic

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

June 2, 2017 3.3 Non-signaling Strategies

classical correlation can be produced by some quantum strategy.

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where clas-

elements as the degree matrix H in a graph G with n vertices and m edges. The graph G is assumed to be connected and the elements of H are assumed to be non-negative. The graph G is assumed to be connected and the elements of H are assumed to be non-negative. The graph G is assumed to be connected and the elements of H are assumed to be non-negative.

BOVM can be defined as the notion of upward simulation in a non-signalling scenario, by using the perfect determinism condition in the definition of the (G, H) condition. In this case, the condition

quantum and non-signalling strategies for the $(\mathfrak{A}, \mathfrak{B})$ -isomorphic signaling, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal

quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that $\text{val}(\Gamma)$ is computable in P^{NP} for both frameworks (Theorem 1.1), and that $\text{val}(\Gamma)$ is P^{NP} -complete for both frameworks (Theorem 1.2).

fractional isomorphism, thus giving the latter an operational interpretation. Second, we show

that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of

Two polynomial systems in non-computing variables, obtained by relaxing the standard integer (and non-negative) fence conditions, is an instance of (A) which might be computationally infeasible.

programming formulations for graph isomorphism so that the same variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally,

we provide a construction for reducing linear binary constraint system games to isomorphism

games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomor-

phic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from

the tensor product and commuting frameworks, are in fact distinct relations, and that the latter

$\{y \in V(G) \cup V(H) \mid \text{for each } x \in V(G) \cup V(H) \text{ for Bob: no path from } x \text{ to } y \text{ as well as for Alice}\}$

POVMS $\mathcal{E}^x = \{E_{y_j}^x : y_j \in V(G) \cup V(H)\}$ for each $x \in V(G) \cup V(H)$ for Alice, and $\mathcal{F}^x = \{F_{y_j}^x : y_j \in V(G) \cup V(H)\}$ for Bob, and $\mathcal{G}^x = \{G_{y_j}^x : y_j \in V(G) \cup V(H)\}$ for Charlie.

A quantum strategy for the (G, H) -isomorphism game consists of a shared entangled state ψ , and

1 Introduction

3.2 Quantum Strategies

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that

$\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are isomorphic, denoted by $G \cong H$. The following is a useful sufficient condition for

say that G and H are *isomorphic* and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory but also

combinatorial optimization, parameterized-complexity, and logic. The *graph isomorphism (GI)*

problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental

Conversely, suppose that $(H(\lambda) \leftarrow G(\lambda) : \delta)$ is an $\mathcal{H}$ -isomorphism of graphs. If H and G are $\mathcal{H}$ -graphs, then G is $\mathcal{H}$ -isomorphic to H and G is easy to

is the unique morphism (11.6) and the unique morphism (11.7) is the unique morphism (11.8).

deterministic strategies. Therefore, classical players can win the (G, H) -isomorphism game only if

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

Consider the (g, h) -entries of the matrices A_G and A_H . We have that D are also obviously nonnegative. We have that D is doubly stochastic. Since the entries where for the first equality we use Lemma 4.1 and for the second one we use (9). The entries of D are also obviously nonnegative. We have that D is doubly stochastic.

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Equation (9). Furthermore, the h^{th} column sum of D is given by:

Define D to be a matrix with rows indexed by $V(G)$ and columns by $V(H)$ such that $D^{gh} = p(h, h|g, g)$, for all $g \in V(G), h \in V(H)$. First, the g^{th} row sum of D is given by $\sum_{h \in V(H)} p(h, h|g, g) = 1$ by $A_G D = D A_H$. We show that D is doubly stochastic and satisfies

Abstract

Lemma 4.2: If G and H are non-signalling isomorphic, then they are fractionally isomorphic. We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the PCSS reduction from inapproximability literature, as well as the CFI construction.

Similarly, we get that

where we use (5) for the first equality, for the second equality we use that p is non-signalling and for the third equality we again use (5). Similarly, we get that

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\varphi: V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are isomorphic. The problem of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism problem* consists of deciding whether two graphs are isomorphic. It is a question with fundamental

Lemma 4.1. Let p be a winning non-signalling correlation for the (G, H) -isomorphism game. Then,

Lemma 4.4. *For any two graphs G and H , if G is isomorphic to H , then G is isomorphic to H .*

Albert Atserias¹, Igor Mamanoski², David E. Roberts³, Robert Špor⁴, Václav Severini⁵, and Antonios Varvitsiotis⁶

Theorem 4.3 ([27]). *Two graphs are fractionally isomorphic if and only if they have a common*

As it turns out, common eigenvalues of G and H are necessary for a common equitable partition.

³University of Cambridge, United Kingdom
⁴University of Cambridge, United Kingdom
⁵University of Cambridge, United Kingdom
⁶University of Cambridge, United Kingdom

for which $C = \{C_1, \dots, C_k\}$ is a partition of $V(G)$ such that for any $i, j \in [k]$, the subgraph induced by $C_i \cup C_j$ is regular, and for any $i \in [k]$, the subgraph induced by C_i is equitable if for any $i \in [k]$, the subgraph induced by C_i is equitable.

We say that C and D have a common equitable partition if there exist equitable partitions containing the edges between C_i and C_j is a semiregular bipartite graph.

Equivalently, a partition $C = \{C_1, \dots, C_k\}$ is equitable if for any $i \in [k]$, the subgraph induced by the vertices in C_i is regular, and for any $i, j \in [k]$, the subgraph induced by the vertices in $C_i \cup C_j$ is equitable.

where each part has size 1. Less trivially, if G is regular, the partition with only one cell is equitable.

Abstract We consider the problem of deciding whether two graphs G and H are isomorphic. A trivial example of an equitable partition is the partition $C = \{V(G)\}$. A non-trivial example of an equitable partition is the partition $C = \{V(G)\}$.

Let G and H be two graphs. We consider the problem of deciding whether G and H are isomorphic. A trivial example of an equitable partition is the partition $C = \{V(G)\}$. A non-trivial example of an equitable partition is the partition $C = \{V(G)\}$.

quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we show that quantum isomorphic graphs are necessarily cospectral.

To show the converse of Lemma 4.2, we use a result of Razborov, Schaefer, and Ullmann [27] which shows that the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we show that quantum isomorphic graphs are necessarily cospectral.

4.2 Fractional isomorphism implies non-signalling isomorphism

On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we show that quantum isomorphic graphs are necessarily cospectral.

□

By Lemma 4.1 we have $p(g, g|h, h) = p(h, h|g, g)$ and $p(h, h|g, g) = p(g, g|h, h)$.

Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact identical.

Our construction is related to the FGLSS reduction from inapproximability literature, as well as the construction of G and H in [10].

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(h)$ if and only if g is adjacent to h . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental

$$\text{Quantum and non-signalling graph isomorphisms} \quad (15)$$

Case 2: If $h^j := x_B \in D_j$ we have that:
 Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

$$\left. \begin{array}{l} \text{if } g \neq h \text{ if } (y, y|g, h) \in C_i \\ \text{if } g \sim h \text{ if } (y, y|g, h) \in C_i \\ \text{if } g = h \text{ if } (y, y|g, h) \in C_i \end{array} \right\} \quad (14)$$

¹University of Bristol, United Kingdom
²University College London, United Kingdom
³University of Bristol, United Kingdom
⁴University of Bristol, United Kingdom
⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China
⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

the proof in two subcases.
 $h := y_A \in D_i$ (the case $x_A \in D_i \& y_A \in C_i$ follows similarly). For clarity of exposition we divide
 definition that $\sum_{y_B} p(y_A, y_B | x_A, x_B) = 0$. It remains to consider the case $g := x_A \in C_i$ and
 Now we proceed to prove (13). If $x_A \in C_i \& y_A \notin D_i$ (or $x_A \in D_i \& y_A \notin C_i$) we have by
 by definition that the correlation p is non-signalling.
Abstract
 and similarly when Alice and Bob exchanged. As this does not depend on the choice of x_B it follows

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where clas-
 sical players can win with certainty if and only if the graphs G and H are isomorphic. We
 then define the notions of quantum and non-signalling isomorphism, by considering perfect
 quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the
 quantum case, we consider both the tensor product and commuting frameworks for nonlocal
 games. We prove that non-signalling isomorphism coincides with the well-studied notion of
 fractional isomorphism, thus giving the latter an operational interpretation. Second, we show
 that, in both tensor product and commuting frameworks, (G, H) -isomorphism is equivalent to an isomorphism
 between two dual systems of noncommuting variables, obtained by dequantizing the standard isomorphism
 program formulation for graph isomorphism. Hence, noncommuting variables (on the basis of this
 correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally,
 we provide a construction for reduced linear games for graph isomorphism. This allows us to produce quantum isomorphic graphs that are nevertheless not isomor-
 phic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from
 the tensor product and commuting frameworks, are in fact distinct relations, and that the latter
 is undecidable. Our construction is related to the GLSS reduction from inapproximability
 literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi: V(G) \rightarrow V(H)$ such that
 $\varphi(g)$ is adjacent to $\varphi(h)$ if and only if g is adjacent to h . When such an isomorphism exists, we
 say that G and H are *isomorphic* and write $G \cong H$. The notion of isomorphism is central to the
 broad area of mathematical research encompassing algebraic and structural graph theory, but also
 combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism (GI)*
 problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental
 importance in computer science. Also define $C_i := \{D_1, \dots, D_k\}$ and $\mathcal{D} = \{D_1, \dots, D_k\}$ and $\mathcal{C} = \{C_1, \dots, C_k\}$
 where D_i is the common partition numbers. Let $n_i = |C_i|$ for all i and let c_{ij}

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mancinska², David E. Robertson³, Robert Samard⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

$$\tilde{\psi} = U^A \otimes U^B \psi, \quad E_{xy}^A = U^A E_{xy}^B U^A, \quad x, y \in V^A \cup V^B \quad (17)$$

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Recall that a correlation $p(y, x' | x, y)$ is quantum if it can be generated by a quantum strategy, i.e., if there exists a quantum state $\psi \in \mathbb{C}^d \otimes \mathbb{C}^d$ and measurements $\{E_{xy}\}_y$ and $\{F_{x'y'}\}_{x'}$ such that

5.1 Common techniques for analyzing quantum strategies

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We

then define the notions of quantum and non-signalling isomorphism, by considering perfect, quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal

games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of a linear programming formulation for graph isomorphism. Similarly, in the commuting framework, quantum isomorphism is equivalent to the feasibility of a semidefinite programming formulation for graph isomorphism.

It is known that the quantum isomorphism game is undecidable in general. In this paper, we

provide a construction for undecidable linear and binary constraint systems for isomorphism games. This allows us to produce quantum isomorphism graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability of graph isomorphism to non-extendability of graph homomorphisms. It is similar to the construction in [6], but it is simpler and more general. It seems likely that our construction is related to the FGLSS reduction from inapproximability of graph isomorphism to non-extendability of graph homomorphisms. It is similar to the construction in [6], but it is simpler and more general. It seems likely that our construction is related to the FGLSS reduction from inapproximability of graph isomorphism to non-extendability of graph homomorphisms.

As mentioned above, a common example of graphs that are fractionally isomorphic but not isomorphic is any pair of non-isomorphic d -regular graphs on n vertices.

1 Introduction

Theorem 4.5. For any graphs G and H we have that $G \cong_f H$ if and only if $G \cong_{ns} H$.

Combining Lemma 4.2 with Lemma 4.4 we immediately get Result 1. Given graphs G and H , an isomorphism from G to H is a bijection $\phi : V(G) \rightarrow V(H)$ such that $\phi(g)$ is adjacent to $\phi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in computer science. In the case $x, y \in V$ (the case $x, y \in E$ is similar). Then, we have that

where the first equality follows from (11), the second one from (12) and the third one by Case 1. Lastly, we show that p is a valid probability distribution. For this, let $x_A, x_B \in V$ and assume that $x_A \in G$ (the case $x_A \in B$ is similar). Then, we have that

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Samardžić⁴, Simone Severini⁵, and Antonios Vavitsiotis⁶

¹Imperial College London, United Kingdom
²University of Bristol, United Kingdom
³University of Oxford, United Kingdom
⁴University of Cambridge, United Kingdom
⁵University of Valencia, Spain
⁶University of Athens, Greece

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism in Hermitian variables. In this paper, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to prove that quantum isomorphism is undecidable. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct and non-equivalent. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI problem.

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism in Hermitian variables. In this paper, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to prove that quantum isomorphism is undecidable. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct and non-equivalent. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI problem.

1 Introduction

It is not hard to see that the vec map is an isometry, i.e., $\text{vec}(A)^\dagger \text{vec}(B) = \text{Tr}(A^\dagger B)$. Also, let $\text{mat} : \mathbb{C}^{d_1 \times d_2} \rightarrow \mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$ be the inverse of the map vec . Given graphs G and H , an isomorphism from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(h)$ if and only if g is adjacent to h . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity and logic. The (G, H) -isomorphism game consists of deciding whether two graphs are isomorphic. It is a question with fundamental consequences of this fact is that the operator $\theta = \text{mat}(u) \text{mat}(u)^\dagger$ is a diagonal matrix with positive entries to ψ , $\{E_{xy}\}_y$ and $\{F_{xy}\}_y$ also generates the correlation $p(y, y|x, x')$ given in (16). Another useful is a valid quantum state with full Schmidt rank. Furthermore, the quantum strategy corresponding

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Vavitsiotis⁶

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Abstract. We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\varphi: V(G) \rightarrow V(H)$ such that $(u, v) \in E(G)$ if and only if $(\varphi(u), \varphi(v)) \in E(H)$. When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The graph isomorphism problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in our field, and it has been one of the most prominent open problems in complexity theory for decades. Since the graph isomorphism game is synchronous, Lemma 5.2 applies to it. However, in our

(v) $p(y, x|x, x') = \psi^\dagger(E^{x_y} \otimes F^{x_{y'}}) \psi = 0$ if and only if $E^{x_y} F^{x_{y'}} = 0$.

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Samal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Abstract

June 2, 2017

isomorphism. First we will need the following definition:

It turns out that one can use Theorem 5.4 to obtain an analogous formulation of quantum graph

PAH for some permutation matrix P , where AG and AH are the adjacency matrices of two graphs.

Recall that an alternative characterization of graph isomorphism is given by the equation $AGP =$

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where clas-

sical players can win with certainty if and only if the graphs G and H are isomorphic. We

then define the notions of quantum and non-signalling isomorphism. Considering perfect

quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the

quantum case, we consider both the tensor product and commuting frameworks for nonlocal

games. We prove that non-signalling isomorphism coincides with the well-studied notion of

fractional isomorphism, thus giving the latter an operational interpretation. Second, we show

that, in the tensor product framework, quantum isomorphism is equivalent to the notion of

fractional isomorphism. On the basis of this

correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally,

we provide a construction for reducing linear binary constraint system games to isomorphism

games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomor-

phic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from

the tensor product and commuting frameworks, are in fact distinct relations, and that the latter

is undecidable. Our construction is related to the

literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\varphi: V(G) \rightarrow V(H)$ such that

$\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we

say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a

broad area of mathematical research encompassing algebraic and structural graph theory, but also

combinatorial optimization, parameterized complexity, and logic. The graph isomorphism

problem consists of deciding whether two graphs are isomorphic. In Section 1.1 we review some

fundamental properties of perfect quantum strategies to the (G, H) -

Quantum and non-signalling graph isomorphism

Albert Atserias¹, Laura Mancinska², David E. Robertson³, Robert Samard⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

$$\sum_{E^{g,h}} = \sum_{(A_G \otimes I_d)(P^{g,h})} = \sum_{(A_G \otimes I_d)(P^{g,h})} \quad (23)$$

²Universitat Politècnica de Catalunya, Spain
³University of Bristol, United Kingdom
⁴University College London, United Kingdom
⁵Charles University, Czech Republic
⁶University College London, United Kingdom and Shanghai Jiao Tong University, China

Abstract
 We introduce a two-player monotone game, called the (G, H) -isomorphism game, where classical players play with Hermitian graphs and quantum players play with Hermitian graphs. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of a quantum semidefinite program. Finally, we show that quantum isomorphism is necessarily stronger than fractional isomorphism. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that the two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

$$(A_G \otimes I_d)P = P(A_H \otimes I_d). \quad (22)$$

Abstract

We introduce a two-player monotone game, called the (G, H) -isomorphism game, where classical players play with Hermitian graphs and quantum players play with Hermitian graphs. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of a quantum semidefinite program. Finally, we show that quantum isomorphism is necessarily stronger than fractional isomorphism. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that the two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

1 Introduction

Conversely, suppose that P is unitary. Since $PP^\dagger = I$ we have that
 Therefore $PP^\dagger = I$ and similarly we have that $P^\dagger P = I$, i.e., P is unitary.
 Given graphs G and H , an isomorphism from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity and logic. The graph isomorphism problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in complexity theory. First, assume that Conditions (i) and (ii) hold. Since the matrices $\{E_{ij}\}_{ij}$ are projectors, and (ii) in the statement of the lemma are equivalent to P being unitary.
Proof. It suffices that show that, assuming the matrices $\{E_{ij}\}_{ij}$ are projectors, Conditions (i)

the following corollary:

Since a projective permutation matrix is unitary, the equation $(A_G \otimes I_d)P = P(A_H \otimes I_d)$ is equivalent to $P^\dagger(A_G \otimes I_d)P = P^\dagger(A_H \otimes I_d)P$, which implies that $A_G \otimes I_d$ and $A_H \otimes I_d$ have the same multiset of eigenvalues. Of course this means that A_G and A_H have the same multiset of eigenvalues, and thus quantum isomorphic graphs are cospectral with respect to their adjacency matrices. Since two graphs are isomorphic if and only if their cospectralities are, we have the following corollary:

corresponds to a quantum isomorphism from a cospectral graph to itself.

Remark 5.9. The above lemma implies that permutation matrices play the role of permutation matrices for quantum isomorphism from a cospectral graph to itself.

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Abstract. We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. We show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. We show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

Taking as a starting point the well-studied notion of fractional isomorphism [26], we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

As a consequence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

As a consequence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

As a consequence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint systems to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

and therefore, combining (24) with (25) it follows that

$$1 \quad \text{Introduction} \quad \left(\sum_{g' : g' \sim g} E_{g'h} \right) = \sum_{g' : g' \sim g} E_{g'h} \quad (25)$$

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi : V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are *isomorphic* and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism problem* consists of deciding whether two graphs are isomorphic. It is a question with fundamental

Conversely, if $(A_G \otimes I_d)P = P(A_H \otimes I_d)$, it follows by (23) that $(A_G \otimes I_d)P = P(A_H \otimes I_d)$ and therefore it follows by (23) that $(A_G \otimes I_d)P = P(A_H \otimes I_d)$.

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

¹*Universitat Politècnica de Catalunya, Spain*
University of Bristol, United Kingdom

University of Bristol, United Kingdom

$$^3 \text{University College London, United Kingdom} \quad (67)$$

^{H 4} Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Note that since quantum commuting strategies for nonlocal games are more general than quantum tensor product strategies, we have that two graphs being quantum isomorphic implies that they are also quantum commuting isomorphic. Similarly, since quantum commuting strategies are

$$(iii) \quad E^{g_h} E^{g_{h'}} = 0, \quad (f|_{\text{re}})(g, g') \neq (h, h')|_{\text{re}}.$$

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our definition of quantum isomorphism is non-trivial on products and commuting frameworks, and that, in contrast to the latter, quantum isomorphism is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an isomorphism from G to H is a bijection $\phi: V(G) \rightarrow V(H)$ such that $\phi(u)$ is adjacent to $\phi(v)$ if and only if u is adjacent to v . When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The problem of isomorphism is a central and broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism (GI)* problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in complexity theory, and it is one of the few problems in the complexity class P that are not known to be in P or NP-complete. The graph isomorphism problem is a central and broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism (GI)* problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental importance in complexity theory, and it is one of the few problems in the complexity class P that are not known to be in P or NP-complete.

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Samal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom, and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players are given vertices $v_1, \dots, v_n$ of the graphs G and H and are asked to output a pair of vertices (u_1, u_2) such that (v_1, u_1) and (v_2, u_2) are edges in G and H respectively. The quantum and non-signalling strategies for the (G, H) -isomorphism game are respectively the quantum and non-signalling homomorphisms from G to H . In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that the (G, H) -isomorphism game coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of restricted polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we prove that quantum isomorphism is undecidable. Finally, we show that the (G, H) -isomorphism game is undecidable for the tensor product and commuting frameworks, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

Motivated by this, in [15] they say that there is a quantum homomorphism from G to H , and write $G \xrightarrow{q} H$ if there exists a perfect quantum strategy for the (G, H) -isomorphism game. Similarly to the isomorphism game, it is not difficult to show that classical homomorphism from G to H , and write $G \xrightarrow{c} H$ if there exists a homomorphism from G to H .

1 Introduction Similarly to the isomorphism game, it is not difficult to show that classical homomorphism from G to H , and write $G \xrightarrow{c} H$ if there exists a homomorphism from G to H . Given graphs G and H and a homomorphism from G to H , it is a function $\varphi: V(G) \rightarrow V(H)$ such that for every vertices $g, h \in V(G)$, if $(g, h) \in E(G)$ then $(\varphi(g), \varphi(h)) \in E(H)$. When such an isomorphism exists, we say that G and H are isomorphic and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity and logic. The graph isomorphism (GI) problem consists of deciding whether two graphs are isomorphic. It is a question with fundamental quantum homomorphisms between each other in both directions. This will imply that quantum homomorphisms must admit we will see that, as in the classical case, any pair of quantum isomorphic graphs must admit of the initial inspiration for the isomorphism game and quantum isomorphisms. In this section

Quantum and non-signalling graph isomorphisms (i) $\mathcal{F}$ is satisfiable

Theorem 6.2. Let $\mathcal{F}$ be a linear BCS with m constraints. Then the following are equivalent:

whether $G_{\mathcal{F}}$ and $G_{\mathcal{F}^*}$ are isomorphic.
 Albert Atserias¹, Laura Mančinska², David E. Roberson³, Robert Šámal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

We are now ready to prove that satisfiability of a linear BCS $\mathcal{F}$ can be reduced to deciding

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

(31)

³University College London, United Kingdom

set in $G_{\mathcal{F}_0}$ of size m . Therefore, with linear BCS $\mathcal{F}$ with m constraints we have

agree on the intersection of their domains $\{i \in [m] \mid \ell_i = f_i^0\}$. This means that the necessary

researchers for this work are: University of London, United Kingdom, and Shanghai Jiao Tong University, China

is a vertex of $\mathcal{F}$ (see [19]). We now show that $G_{\mathcal{F}_0}$ always contains an independent set of size m (and thus $\alpha(G_{\mathcal{F}_0}) = m$).

For each constraint C_i , let f_i^0 denote the zero assignment to the variables in S_i . Note that (ℓ_i, f_i^0)

We now show that $G_{\mathcal{F}_0}$ always contains an independent set of size m (and thus $\alpha(G_{\mathcal{F}_0}) = m$).

$G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ have the same number of vertices

homogenization a linear BCS always has a solution, namely the all-zero assignment. Also note that

BCS obtained from $\mathcal{F}$ by changing the right-hand sides of all of the constraints to 0. Note that the

Given any linear BCS $\mathcal{F}$, we define the *homogenization* of $\mathcal{F}$, denoted by $\mathcal{F}_0$, to be the linear

each constraint C_i of $\mathcal{F}$ is replaced by $C_i(x_i) = 0$. We introduce a new player u_0 and add the constraint

for any $i \in [m]$, $\ell_i = f_i^0$. We then define the notions of quantum and non-signalling isomorphism, by considering perfect

Note that the vertices of $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ are in one-to-one correspondence. In the

quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the

the FGLSS reduction from [27], which is well-studied in the literature.

there exists $x_i \in S_i$ such that $f_i^0(x_i) = 0$. We remark that such a construction is related to

Furthermore, we find an edge between two vertices (i, j) if and only if $\ell_i = f_i^0$ and $\ell_j = f_j^0$.

For each constraint C_i , and each basis b_i of S_i , we find a vertex v_i in $G_{\mathcal{F}_0}$ such that $\ell_i(v_i) = 0$ and $f_i^0(v_i) = 0$.

To any linear BCS $\mathcal{F}$, we find a vertex v_i in $G_{\mathcal{F}_0}$ such that $\ell_i(v_i) = 0$ and $f_i^0(v_i) = 0$.

graph isomorphism. As first shown by [27], we can reduce the problem of deciding whether a

In this section we prove that quantum isomorphism games are necessarily coNP-complete. Finally,

we provide a construction for reducing linear binary constraint system games to isomorphism

games. This allows us to produce quantum isomorphic graphs that are never isomorphic.

Furthermore, it allows us to show that our two notions of quantum isomorphism, from

the tensor product and commuting frameworks, are in fact distinct relations, and that the latter

BCS that is quantum satisfiable but not satisfiable from any non-player graph

is undecidable. Our construction is related to the FGLSS reduction from inapproximability

literature, as well as the CFL construction. One can also use a result of Arkhipov [1] to construct a linear

the example given above in (30) corresponds to the Mermín-Péres magic square game [19] which

There are many classes of linear BCS's that are quantum satisfiable but not satisfiable. Indeed,

version.

game differently is simply because the reduction to quantum isomorphism is more natural for this

of the two versions of the BCS game follow from the results in [10]. The reason we define the

satisfiability is the same as the case of quantum commuting strategies, the equality of ϕ

the BCS game if and only if one exists for their. Moreover, this implies that a perfect quantum strategy exists for a

broad area of mathematical research encompassing algebraic and structural graph theory.

assess by [27] and from this it easily follows that a perfect quantum strategy exists for a

combinatorial optimization parameterized by the complexity of the graph. The graph

1 Introduction

[illegible]

□ quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show

However, if there exists $x \in S_{\text{EGLSS}}$ such that $f(x) \neq f(x)$, then the decision problem EGLSS is undecidable. Our construction is related to the EGLSS reduction from inapproximability literature as well as the CFL construction.

how that the correlation defined in (35) evaluates to zero whenever the winning conditions of the BCS game are not fulfilled (recall Equation (4)). Now, by construction of the

on her half of ψ_d to obtain an assignment $f : S_\ell \rightarrow \mathbb{F}_2$ that satisfies C_ℓ . Upon receiving constraint C_k , Bob acts similarly, except that he performs the measurement $\{E_{\text{T}}^{(k,f)} : f \text{ satisfies } C_\ell\}$ to get an

Quantum and non-signalling graph isomorphisms

Albert Atserias¹, Laura Mančinská², David N. Rubinfeld³, Robert Samal⁴, Simone Severini⁵, and Antonios Varvitsiotis⁶

$$0 = 6x + 9x + 8x$$

$$0 = 8x + 5x + 7x$$

$$0 = 4x + 7x + 1x$$

¹Universitat Politècnica de Catalunya, Spain

²University of Bristol, United Kingdom

³University College London, United Kingdom

⁴Charles University, Czech Republic

⁵University College London, United Kingdom and Shanghai Jiao-Tong University, China

⁶Nanyang Technological University and Centre for Quantum Technologies, Singapore

June 2, 2017

Abstract

We introduce a two-player nonlocal game, called the (G, H) -isomorphism game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, by considering perfect quantum and non-signalling strategies for the (G, H) -isomorphism game, respectively. In the quantum case, we consider both the tensor product and commuting frameworks for nonlocal games. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the latter an operational interpretation. Second, we show that, in the tensor product framework, quantum isomorphism is equivalent to the feasibility of two polynomial systems in non-commuting variables, obtained by relaxing the standard integer programming formulations for graph isomorphism to Hermitian variables. On the basis of this correspondence, we show that quantum isomorphic graphs are necessarily cospectral. Finally, we provide a construction for reducing linear binary constraint system games to isomorphism games. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two notions of quantum isomorphism, from the tensor product and commuting frameworks, are in fact distinct relations, and that the latter is undecidable. Our construction is related to the FGLSS reduction from inapproximability literature, as well as the CFI construction.

1 Introduction

Given graphs G and H , an *isomorphism* from G to H is a bijection $\varphi: V(G) \rightarrow V(H)$ such that $\varphi(g)$ is adjacent to $\varphi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are *isomorphic* and write $G \cong H$. The notion of isomorphism is central to a broad area of mathematical research encompassing algebraic and structural graph theory, but also combinatorial optimization, parameterized complexity, and logic. The *graph isomorphism (GI) problem* consists of deciding whether two graphs are isomorphic. It is a question with fundamental

$$0 = 6x + 8x + 4x$$

$$0 = 9x + 5x + 7x$$

$$0 = 8x + 2x + 1x$$

- [4] László Babai. Automorphism groups, isomorphism, reconstruction. In *Handbook of Combinatorics (Vol. 2)*, pages 1447–1540, 1995. doi:2332228.2332236.
- [5] László Babai. Graph isomorphism in quasipolynomial time. 2015. arXiv:1512.03547.
- [6] László Babai, Eugene M. Luks, and Gábor L. Miller. Graph isomorphism in polynomial time for bounded degree graphs. *Journal of Combinatorial Theory, Series B*, 118:228–267, 2016. arXiv:1212.1724.
- [7] Nicolas Brümmer, Daniel Cavalcanti, Stefano Pironi, Valerio Scarani, and Stephanie Wehner. Bell nonlocality. *Reviews of Modern Physics*, 86:419–478, 2014. arXiv:1303.2849.
- [8] Jin-Yi Cai, Martin Fürer, and Neil Immerman. An optimal lower bound on the number of variables for graph identification. *Combinatorica*, 12(4):389–410, 1992. doi:10.1007/BF01305232.
- [9] Peter J. Cameron, Ashley Montanaro, Michael W. Newman, Simone Severini, and Andreas Winter. On the quantum complexity of graph isomorphism. *Journal of Combinatorial Theory, Series B*, 118:228–267, 2016. arXiv:1608.02728.
- [10] Richard Cleve, David Longshor, and William Stockmeyer. Lower bounds for quantum communication complexity. *SIAM Journal on Computing*, 23(6):1970–1994, 1994. doi:10.1137/S0036141094268080.
- [11] Richard Cleve and John Watrous. Quantum communication complexity of graph isomorphism. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing*, pages 91–100, 2009. doi:10.1145/1555554.1555572.
- [12] Ueli Feige and Shmuel Safra. A randomized algorithm for finding small dense subgraphs. In *Proceedings of the 27th Annual ACM Symposium on Theory of Computing*, pages 137–147, 1995. doi:10.1145/358906.358922.
- [13] Martin Grohe. *Descriptive Complexity*. Cambridge University Press, 2013. Available online. Draft manuscript, 2013.
- [14] Martin Grohe and Martin Otto. Pebble games and linear equations. *Journal of Symbolic Logic*, 80(03), 2012. arXiv:1204.1990.
- [15] Laura Mančinska and David E. Roberson. Quantum homomorphisms. *Journal of Combinatorial Theory, Series B*, 118:228–267, 2016. arXiv:1212.1724.
- [16] Nanyang Technological University and Centre for Quantum Technologies, Singapore
- [17] Laura Mančinska, David E. Roberson, and Antonios Varvitsiotis. Semidefinite relaxations of quantum isomorphism. Available online, 2016.
- [18] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism. II. *Journal of Symbolic Computation*, 60:991–1127, 2014. arXiv:1301.1493.
- [19] Nathaniel D. Susskind. Entanglement for the Hamiltonian method and related problems. *Physical Review Letters*, 65:3373–3376, 1990. doi:10.1103/PhysRevLett.65.3373.

Abstract

We introduce a two-player nonlocal game, called the (H, G) -pebble game, where classical players can win with certainty if and only if the graphs G and H are isomorphic. We then define the notions of quantum and non-signalling isomorphism, and prove that the quantum and non-signalling games for the (H, G) -pebble game, respectively, coincide with the quantum and non-signalling games for the (H, G) -pebble game. We prove that non-signalling isomorphism coincides with the well-studied notion of fractional isomorphism, thus giving the first non-trivial lower bound for the quantum complexity of the (H, G) -pebble game. We also show that the quantum complexity of the (H, G) -pebble game is at least $\Omega(\log |V(H)|)$. On the basis of this correspondence, we show that quantum isomorphism is equivalent to the existence of a quantum communication protocol for the (H, G) -pebble game. This allows us to produce quantum isomorphic graphs that are nevertheless not isomorphic. Furthermore, it allows us to show that our two-player game is not in P. We also show that the quantum complexity of the (H, G) -pebble game is at least $\Omega(\log |V(H)|)$. Our construction is related to the CFI construction of Immerman, as well as the CFI construction.

1 Introduction

Given graphs G and H , we say that G and H are isomorphic if there exists a bijection ϕ from $V(G)$ to $V(H)$ such that $\phi(g)$ is adjacent to $\phi(g')$ if and only if g is adjacent to g' . When such an isomorphism exists, we say that G and H are isomorphic. The graph isomorphism problem (GI) is the problem of deciding whether two graphs are isomorphic. It is a question with fundamental importance in combinatorial optimization, parameterized complexity, and logic. The graph isomorphism problem (GI) is one of the most important open problems in complexity theory. It is a question with fundamental importance in combinatorial optimization, parameterized complexity, and logic. The graph isomorphism problem (GI) is one of the most important open problems in complexity theory.

[20] Benjamin Musto and Jamie Vicary. Quantum Latin squares and unitary error bases. 2015.